Date: 02 July 2026	**Team ID:** SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	**Maximum Marks:** 2 Marks

DATA FLOW DIAGRAM TEMPLATE

Data Flow Diagram (DFD) Legend

Create a DFD to show how data moves through your system - between external entities, processes, and data stores.

Symbol Shape	DFD Element Name	Description / Purpose in System
Oval / Rounded Shape	External Entity	A person, organization, or external system that sends or receives data (e.g., Credit Officer, Applicant).
Rectangle with Numbered Header	Process	An activity that transforms incoming data into outgoing data (e.g., 1.0 Validate Input, 2.0 Evaluate Policies).
Rectangle (no number, solid borders)	Data Store	A repository where data is held for later use (e.g., Model Pickle Files, CSV historical dataset).
Labeled Arrow	Data Flow	The movement of data between elements, labeled with the specific variables being passed.

Level 0 DFD (Context Diagram)

The context diagram illustrates the system boundaries. The credit card applicant inputs 12 demographic and financial attributes, and the underwriting system outputs the decision (Approved/Rejected) along with risk probabilities and execution logs.

Level 1 DFD (Detailed Process Flows)

The detailed flow decomposes the underwriting pipeline into four sequential processes:

Process ID	Source Element	Process Description	Data Store / Destination
1.0	home.html form	Sanitizes fields and validates data boundaries (e.g., positive values for income/age).	Transfers cleaned variables to Process 2.0.
2.0	Process 1.0 data	Evaluates the 4 hard-coded banking policies. If rules breach, short-circuits with rejection.	Audit database / result.html (Immediate Reject).
3.0	Process 2.0 pass	Loads scaler.pkl, standardizes continuous features, and loads credit_model.pkl (Random Forest).	Passes default probability vector to Process 4.0.
4.0	Process 3.0 probability	Compares predictions against threshold.pkl. Determines final outcome.	Renders green/red outcomes panel on result.html.